PINION GEAR/RING GEAR/TONE RING (Continued)

(10) Check bearing preload torque with an inch pound torque wrench (Fig. 100). The torque necessary to rotate the pinion gear should be:

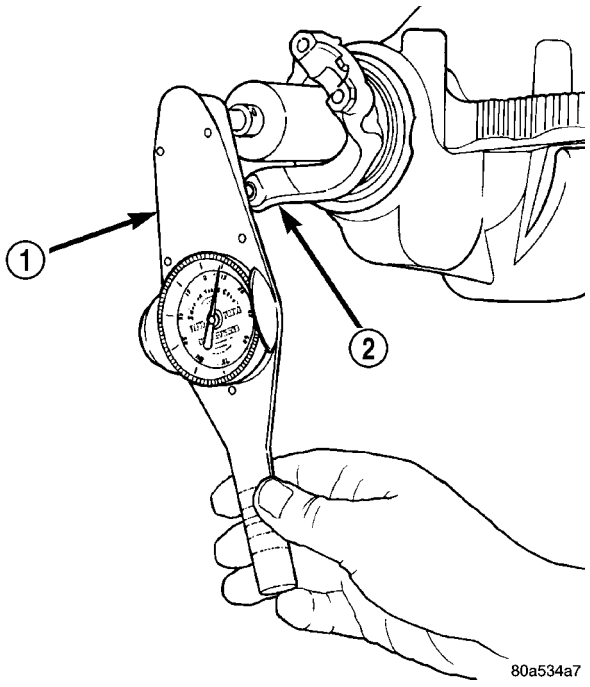

Fig. 100 PINION ROTATING TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

- Original Bearings: 1 to 2.26 N·m (10 to 20 in. lbs.).
- New Bearings: 2.26 to 4.52 N·m (20 to 40 in. lbs.).

(11) If rotating torque is above the desired amount, remove the pinion yoke and increase the preload shim pack thickness. Increasing the shim pack thickness 0.025 mm (0.001 in.) will decrease the rotating torque approximately 0.9 N·m (8 in. lbs.).

(12) If the maximum tightening torque is reached prior to achieving the desired rotating torque, remove the pinion yoke and decrease the thickness of the preload shim pack. Decreasing the shim pack thickness 0.025 mm (0.001 in.) will increase the rotating torque approximately 0.9 N·m (8 in. lbs.).

(13) Invert differential case and start two ring gear bolts. This will provide case-to-ring gear bolt hole alignment.

CAUTION: Never reuse ring gear bolts, the bolts can fracture causing extensive damage. Failure to heed caution may result in damage.

(14) Invert differential case in the vise.

(15) Install **new** ring gear bolts and alternately tighten to 108 N·m (100 ft. lbs.) (Fig. 101).

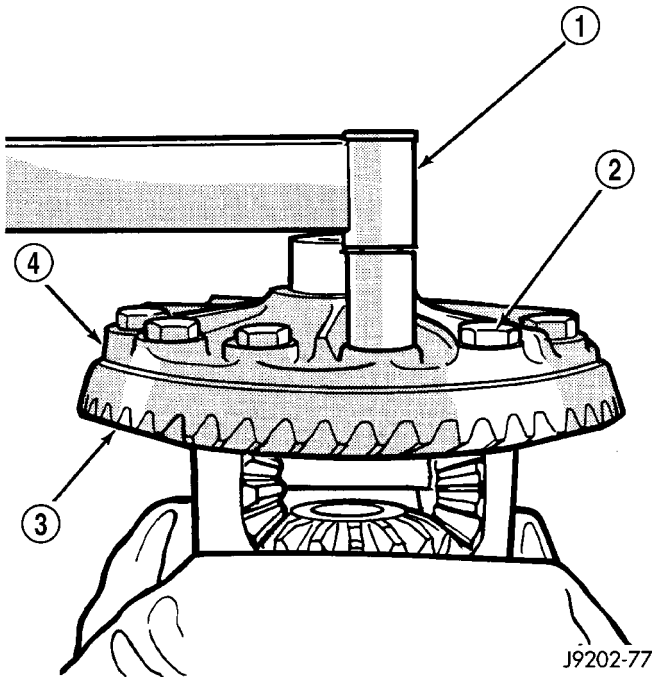

Fig. 101 RING GEAR

1 - TORQUE WRENCH
2 - RING GEAR BOLT
3 - RING GEAR
4 - CASE

(16) Install differential in axle housing and verify gear mesh and contact pattern. Refer to Adjustment (Gear Contact Pattern).